

Desktop Organisms: Coherent Character Animation and Steerable Behavior Manifolds in Neural Cellular Automata, in Two and Three Dimensions

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Working draft — single-seed experiments, no ablations yet.

Abstract

Neural cellular automata (NCA) grow persistent, self-repairing images from a single seed, but prior work animates them only as static attractors, dynamic textures, or physics-driven soft bodies. We show that an NCA can perform *coherent character animation as a limit cycle of its own dynamics*: a phase vector conditioned on every cell sweeps a closed loop through sprite-space, and the automaton learns to *track* the moving target, yielding temporally coherent talking, blinking, and locomotion cycles with no stored frames. We extend the recipe in three directions. (1) *Behavior manifolds*: a 10-factor latent z , applied by bounded FiLM modulation and trained on a procedurally generated corpus of 2,048 animation cycles, yields a single 68 KB network whose moods (sleep, dread, mania, ...) are regions of a continuous space, steerable at runtime by any external process — including a large language model. (2) *Volumetric organisms*: the same machinery lifted to 32^3 voxel grids produces, to our knowledge, the first real-time, interactively rendered 3D NCA performing a learned gait cycle. (3) *A deployed system*: all creatures run at 60 automaton steps/s as transparent-window macOS “desktop pets,” trained in PyTorch and executed by runtime-generated Metal kernels under a byte-level weight contract. We report the practical findings that made this work: integer-harmonic loop closure in target authoring, tanh-bounded FiLM gain (unbounded gain reliably diverges), and gradient-checkpointed backpropagation through time for volumetric training. Total weights for six organisms: under 260 KB.

1 Introduction

Desktop companions have historically been spritesheets: fixed frames, fixed behaviors. We asked whether a companion could instead be a *dynamical system* — an organism whose body is a neural cellular automaton [1], grown from a seed, healing when damaged, and whose animation is not playback but ongoing physics.

The obstacle is temporal coherence. Generative models that synthesize frames independently must re-establish identity every frame; NCAs invert the problem, because state persists and updates are local and small — but prior NCAs converge to *fixed points* (static images), not *orbits* (animations). Our central move is to replace the fixed-point attractor with a target that moves

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along a closed loop, conditioned on a phase clock $(\sin \theta, \cos \theta)$ broadcast to every cell, with θ advancing *during training rollouts*. The automaton therefore learns tracking dynamics: how to be mid-transition between poses, not merely how to render poses.

Contributions, in the order they were built and verified:

1. **Phase-conditioned limit-cycle NCA** for character animation (talking/blinking face; walking creature), including *transition training*: behavior flags are re-rolled mid-life during training, so behavior switches are learned conversions of an existing body rather than regrowths.
2. **Bounded-FiLM behavior manifolds**: a factored latent $z \in [0, 1]^{10}$ (motion amplitude, droop, gaze, tremor, walkness, ...) modulates the update rule's hidden layer, trained over a procedurally generated corpus densely sampling the factor box, with named anchors and mid-life z re-sampling. Novel behavior is then *interpolation between trained factors* rather than extrapolation between behavior islands. A file-based control channel lets external processes (including an LLM in an adjacent terminal) steer z live.
3. **Volumetric cyclic NCA**: perception lifted to identity + Sobel $_{x,y,z}$ over 32^3 grids, trained with gradient-checkpointed BPTT and rendered by an emission-absorption raymarcher with gradient shading, at interactive rates on a laptop GPU. The creature performs a learned azimuthal tentacle-wave gait; damage is a sphere, and regeneration is volumetric.
4. **Engineering results** we believe transfer: (a) all time-frequencies in authored cycle targets must be integer harmonics of the base period or the loop does not close and the NCA faithfully learns the seam-lurch; (b) unbounded FiLM gain $(1 + \gamma)$ on a residual dynamical system diverges reliably (ours at iteration 19.5k, poisoning the sample pool); tanh-bounding γ , a $10\times$ lower conditioning learning rate, and discarding non-finite batches eliminated the failure; (c) a byte-level weight format contract enables training in PyTorch and inference in runtime-compiled Metal with visual-equivalence verification.

2 Related Work

Growing NCA. Mordvintsev et al. [1] established the recipe we build on: 16-channel state, fixed identity/Sobel perception, residual update with stochastic firing, alive masking, and pool training with damage for persistence and regeneration. Their targets are static; ours orbit.

Dynamic textures. Self-organising textures [2] and DyNCA [3] produce motion, including flow-controlled dynamic texture, but the object of synthesis is texture statistics rather than an articulated character with persistent identity and discrete behaviors.

Conditioned and multi-target NCA. Prior work conditions NCAs to grow multiple targets from one network. Our manifold differs in that the latent is *factored over motion parameters* and densely sampled in training, making interpolation semantically meaningful, and in that conditioning enters by (bounded) FiLM modulation rather than appended channels.

3D NCA. Sudhakaran et al. [4] grow static Minecraft artefacts; Horibe et al. [5] grow and regenerate soft robots whose locomotion arises from a physics simulator acting on the grown morphology; Mesh NCA [6] runs on surface manifolds. We are not aware of prior work in which a volumetric NCA *performs* a learned animation cycle in its own state space and is rendered interactively; we claim this narrowly and welcome correction.

3 Method

3.1 Base organism

We use the standard Growing-NCA cell: state $\mathbf{s} \in \mathbb{R}^{16}$ per cell (RGBA + 12 hidden), perception by fixed per-channel kernels (identity, Sobel_x, Sobel_y; +Sobel_z in 3D), a two-layer 1×1 update network (128–192 hidden units, ReLU, zero-initialized output), stochastic per-cell firing ($p=0.5$), and the pre/post alive mask on $\alpha > 0.1$. Training uses a 256–1024 sample pool, worst-sample reseeding, circular (2D) or spherical (3D) damage, rollouts of 48–96 steps, MSE to target on RGBA, per-parameter gradient normalization, and Adam. States are clamped to $[-8, 8]$ (inert for healthy dynamics; blowup insurance).

3.2 Limit cycles: animation as tracking

An animation is authored as $F=12$ frames on a closed loop, indexed by phase $\theta \in [0, 2\pi)$. Every cell receives $(\sin \theta, \cos \theta, b)$, where b is a behavior flag (still/talking; idle/walking). During each training rollout θ advances by $\Omega = 2\pi/240$ per step, and the loss is evaluated at several rollout depths against the *phase-lerped* target $T(b, \theta)$. Because the supervised target moves while the automaton runs, the learned dynamics track continuously; at runtime the cycle can be driven at any speed, held, or reversed. With probability 0.15 a pool sample’s b is flipped mid-life, so behavior transitions are trained as conversions of the current body.

Loop closure. Every time-frequency in the authored frames must be an integer multiple of the base period. A draft of our walking creature used frequencies like 1.3θ ; frame $F-1$ then fails to flow into frame 0, and the NCA dutifully learns a once-per-cycle lurch. We verify closure numerically before training (the seam delta must not exceed adjacent-frame deltas).

3.3 Behavior manifolds: bounded FiLM over a factored corpus

We parameterize the creature’s entire motion style by $z \in [0, 1]^{10}$: walkness, amplitude, wave number, droop, bob, eye openness, gaze, brightness, tremor, splay. A generator renders a closed 12-frame cycle for *any* z ; we sample 2,048 cycles (anchors such as *sleep*, *dread*, *manic* oversampled) as the training corpus. z conditions the network by FiLM: $h \mapsto \text{ReLU}(h \cdot (1 + \tanh \gamma(z)) + \beta(z))$, computed once per step since z is spatially uniform. Phase still enters as channels: the clock is fast, the mood is slow. Mid-life z re-sampling (15%) trains transitions across the manifold. At runtime z eases toward a target at 4%/frame, so moods morph through intermediate bodies; a JSON control file allows external steering, and named anchors give language-level handles (“dread”) that an LLM can emit directly.

Stability. With unbounded gain $(1 + \gamma)$, training diverged at iteration 19,500 of 60,000: a single batch reached non-finite values which then poisoned the sample pool and every subsequent checkpoint (which overwrote the last good one). The restarted run — $\tanh$ -bounded γ , conditioning learning rate 2×10^{-4} vs. 2×10^{-3} base, non-finite batches discarded before optimizer and pool, versioned checkpoints — ran 60k iterations without incident and reached a slightly *better* loss at matched iterations.

3.4 Volumetric organisms

In 3D, perception becomes 4 fixed kernels per channel (identity plus separable Sobels, $\text{smooth}(1, 2, 1) \otimes \text{smooth} \otimes \text{deriv}$ i.e. 64 features; targets are procedural voxel sculpts (64^3 supersampled, box-filtered to 32^3) so the loss is direct voxel MSE — no differentiable renderer. Naïve BPTT through 64 steps at batch 8

Creature	Type	Trained on	Iters	Final MSE	it/s	Weights
Bonsai	2D static	M1 Pro (MPS)	8,000	4.7×10^{-4}	4.0	33 KB
Lain	2D cyclic, 2 behaviors	M1 Pro (MPS)	12,000	8.5×10^{-4}	4.0	34 KB
Shoggoth Mk. I	2D cyclic, walk	M1 Pro (MPS)	24,000	8.1×10^{-3}	4.2	34 KB
Shoggoth Mk. II	2D manifold, $z \in \mathbb{R}^{10}$	RTX 4090	60,000	9.5×10^{-3}	9.1	68 KB
Bonsai 3D	32^3 static	RTX 4090	2,000 [†]	3.4×10^{-3}	3.3	41 KB
Shoggoth Mk. III	32^3 cyclic, gait	RTX 4090	24,000	2.2×10^{-4}	2.7	43 KB

Table 1: The menagerie. it/s = training iterations per second (the 4090 was shared with unrelated jobs; rollouts are launch-latency-bound, not compute-bound — CUDA graphs are an obvious next optimization). MSE scales are not comparable across rows (image vs. volume; static vs. moving target). [†]validation run, truncated; finishing run in progress.

stores ~ 14 GB of activations; checkpointing rollouts in 8-step segments reduces peak memory $\sim 8\times$ for $\sim 35\%$ recompute. Display is an emission-absorption raymarcher (trilinear sampling, transmittance early-out, diffuse shading from the density gradient) in a Metal compute kernel; a 32^3 volume at 512×512 rays is negligible load for an Apple M1 Pro. Damage is a sphere; clicks are resolved by marching the pick ray on the CPU against the shared-memory state buffer. The Mk. III creature’s walking gait is an azimuthal traveling wave through a ring of eight tentacles — a motion with no 2D equivalent.

3.5 System

Training (PyTorch, CUDA/MPS) and inference (Swift, runtime-generated Metal source) share a family of flat little-endian weight formats (NCA1/2/3, NC3D/NC3C; 33–68 KB per creature). The kernel source is generated per creature with conditioning width, hidden width, and FiLM baked in as constants. Equivalence is enforced by contract — perception ordering, padding, alive-mask semantics, firing semantics, clamping — and verified visually: weights trained in PyTorch must grow the same organism through the Metal runtime from a single seed (Fig. 1). The deployed application hosts each organism in a transparent, borderless, always-on-top window; weights hot-reload every 2 s, which during training lets the user watch the organism improve — and occasionally molt, since a body grown by an earlier checkpoint is out-of-distribution for a later rule.

4 Results

Qualitatively (Figs. 2–4): identity is stable across cycles because it is stored in persistent state, not re-synthesized; animated regions are exactly where the learned rule is phase-sensitive. Behavior switches (still→talking, idle→walking) proceed by local conversion of the existing body, as trained. Damage anywhere — including spherical wounds through the 3D organism — regrows, and regrowth resynchronizes with the ongoing cycle, which falls out of the pool containing damaged mid-phase samples. Runtime cost on an M1 Pro laptop is ~ 60 automaton steps/s per creature with rendering, at trivial GPU load in 2D and modest load in 3D.

5 Limitations and Honest Framing

The phase clock is *external*: these organisms track a conditioning oscillator rather than keeping time themselves. Two fine-tuning experiments that remove the clock give a characterized partial



Figure 1: **Left:** procedural 2D target. **Center:** organism grown by the Metal runtime from a single seed (8k iters, MPS-trained). **Right:** the volumetric bonsai raymarched from a 25° orbit after only 500 iterations of 3D training — the same tree, now with an interior.



Figure 2: Limit-cycle animation through the Metal runtime (frames from single continuous rollouts). **Left:** face creature mid talk-cycle; identity (hair, clip, gaze) is carried by persistent state while mouth cells track the moving target. **Right:** Mk. I walk gait: a traveling wave through the bottom tentacles.

result. Naïvely deleting the phase inputs collapses the gait to its *temporal mean* (a motionless blur of all phases) — the cheapest MSE response to a schedule the automaton cannot stay synchronized with. Adding a synchronization curriculum (rollout horizons growing $12 \rightarrow 96$ steps) and *oscillator distillation* (an auxiliary loss asking the alive-population mean of two hidden channels to encode $\cos \theta, \sin \theta$) prevents collapse: the clockless organism then moves vigorously and indefinitely with intact identity (consecutive-frame motion $6\times$ the collapsed baseline). But recurrence analysis over 600-step rollouts shows no stable period — only a weak local minimum at the trained period — indicating the distributed phase code decoheres, plausibly because stochastic per-cell firing injects phase noise faster than the learned coupling re-synchronizes it. The automaton retains *that* it moves, not *the* cycle. Stronger coupling mechanisms (explicit neighbor phase terms, reduced firing noise, longer curricula) are the obvious next experiments. MSE training softens precise geometry — fine for protoplasm, hostile to machined shapes; edge-weighted or perceptual losses are untested here. All results are single-seed, without ablations or quantitative animation metrics (a cycle-consistency measure — state distance to itself one period later — would be the natural one). The manifold’s interpolation quality is demonstrated, not measured. The 3D first-of-kind claim is scoped to real-time interactive rendering of a learned in-state gait; adjacent work exists in grown artefacts and physics-driven soft robots.

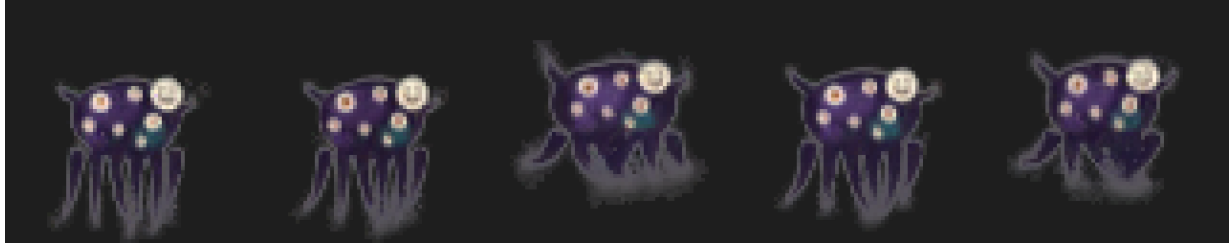


Figure 3: One 68KB network, five moods: *sleep*, *dread*, *manic*, *content*, *walk* anchors of the 10-factor manifold, each grown from seed under a different z . At runtime the organism morphs continuously between such points; external processes steer z through a control file.



Figure 4: Shoggoth Mk. III: a 32^3 volumetric NCA performing its churn gait while the camera orbits (single continuous rollout; frames left-to-right, top-to-bottom). Different surface eyes rotate into view — the organism has an interior and a back.

6 Future Work

Internalizing the clock (in progress); projecting LLM activity — conversation embeddings, thinking-trace summaries — into the behavior manifold, making the organism an ambient display of an agent’s cognition; merging the manifold and volumetric lines (FiLM in NC3C); 64^3 grids with edge-aware losses for rigid characters; CUDA-graph rollout capture, which we expect to dominate all hardware choices at these model sizes.

Acknowledgments

The organisms were trained on the first author’s machines (M1 Pro laptop; a shared RTX 4090 on the LAN) over roughly one day of collaboration between the authors. The bonsai is doing fine. The shoggoths are doing better.

References

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